

**Multidimensional Influences on Reaction Time and Model
Comparison: An Empirical Analysis of Stepwise AIC, Stepwise BIC,
Lasso, and Ridge Regression**

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Abstract

We were curious about what affects how quickly people react, so we did a study using survey data from our STAT425 class. We averaged those scores, as it seemed like the fairest way to compare across people. Our main research questions is: Among all the self-reported behaviors from college students, which survey variables can accurately predict their average reaction time? How consistent are different modeling methods (like stepwise regression with AIC/BIC, Lasso, and Ridge) in terms of which variables they pick? Once the data was cleaned up—some options were too detailed or overlapped—we grouped them a bit and moved on to modeling. We tried a few common ones: AIC and BIC stepwise regression, Lasso, and Ridge. The goal was simple—figure out which one could best predict the reaction time. Ridge ended up doing the best, at least for our dataset. The AIC model followed pretty closely. Lasso and BIC didn't perform as well.

Introduction: Background and Research Question

Reaction time means how long it takes a person to respond after seeing or hearing a stimulus. It matters a lot in real life — like how fast someone reacts while driving, how quick they respond in video games, or even just to check if someone is focused or distracted. Many different things can affect reaction time, such as sleep quality, whether they're being interrupted, the type of device they're using, or even how noisy their environment is. While some studies have looked into one or two of these factors, very few have tried to consider daily habits, mental state, and surroundings all together.

In this project, we used a survey dataset that recorded participants' daily routines, whether they get distracted easily, what devices they use, their emotional state, and more. We also collected the time it took for them to complete three reaction tests, and used the average of those three as their final reaction time. The goal is to figure out which of these variables actually have a significant impact on how fast someone reacts.

To answer our research question, we used several statistical models and performed both model fitting and diagnostic checks on the full dataset. The goal isn't to predict new data, but to understand how well each model explains the variation in reaction time in the current sample, and to explore the structure of these influencing factors. Through this, we hope to better understand what affects individual reaction speed and provide useful insights for future studies on cognitive performance or behavior modeling.

Exploratory Data Analysis

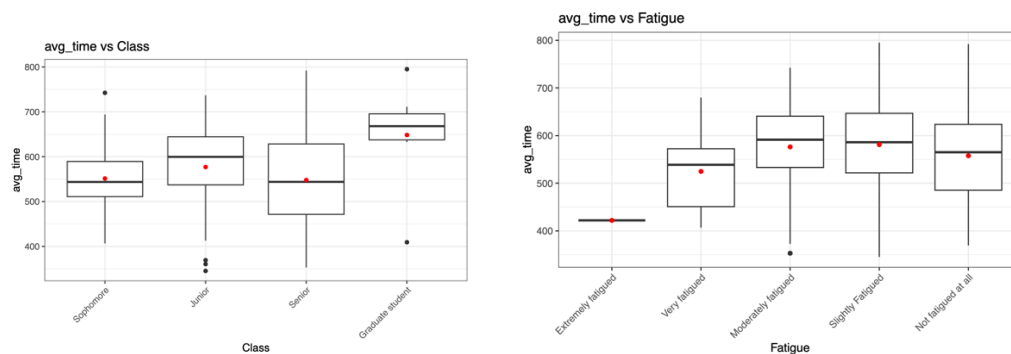
The study includes 145 samples and covers a wide range of variables across several dimensions, including

1. demographic information: Age, Class
2. lifestyle habits : AvgSleepTime, LastNightSleep, CaffeinIntake, alcoholeIntake, AvgHoursExercise
3. psychological states: Fatigue, Stress, Cautious, VisualAcuity.
4. technology usage: InputDevice, UsePrimaryHand, PrimarilyHand, RefreshRate, DeviceOS,
5. environmental factors: Noise_Level, Distraction, Temperature, WiFi.

To better understand what affects average reaction time (avg_time, measured in milliseconds), we started with some exploratory analysis on the cleaned survey data.

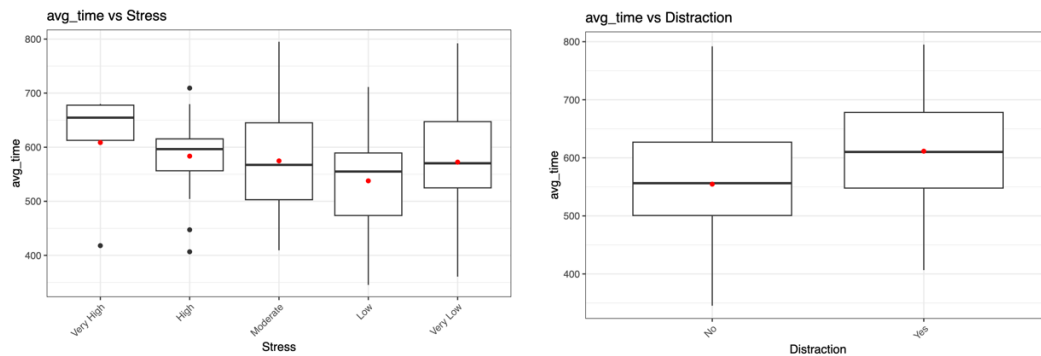
We visualized the distributions of continuous variables like AvgSleepTime and Age, and used boxplots to compare reaction time across different groups. For example, we noticed a clear trend: participants who reported having “low fatigue” generally had faster reaction times than those in the “high fatigue” group. Also, students who played video games often or exercised regularly tended to respond more quickly.

We also found that the distribution of avg_time was a bit skewed and included a few extreme values. However, since these differences likely reflect natural variation among individuals, we decided not to remove them.



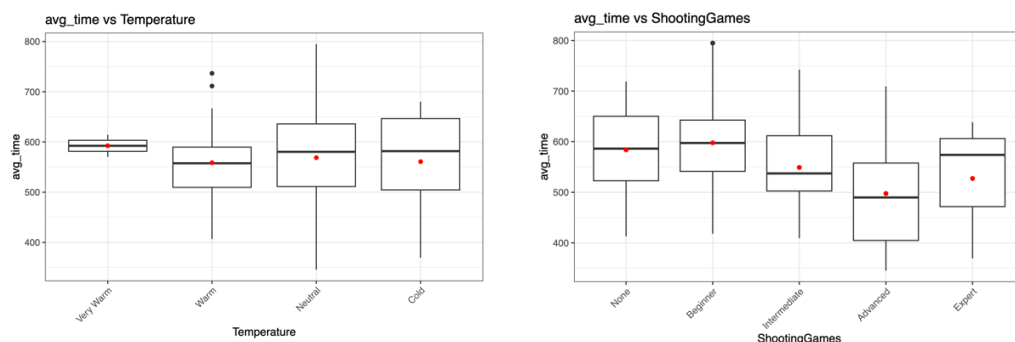
For the left graph, differences in reaction time across school years might be related to factors like academic stress, irregular sleep schedules, or changes in workload. Graduate students usually face more research responsibilities, and juniors often have the heaviest course load.

When looking at the relationship between perceived fatigue and reaction time, we actually found that those who reported “moderate fatigue” had slower responses. One possible reason is that their fatigue level wasn’t high enough to trigger compensatory behaviors, like making extra effort to stay focused. But once the fatigue become higher, the reaction time would definitely become longer.



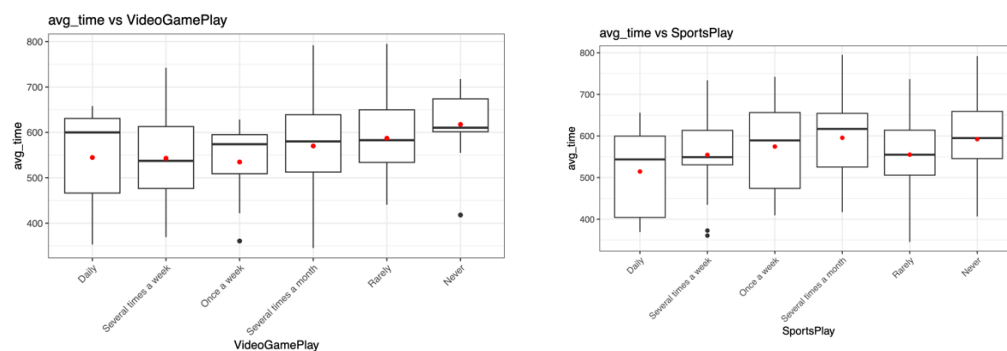
We found that moderate stress might actually help people stay focused and respond faster. But if the stress is too high or too low, it could lead to distractions or a lack of motivation, which slows down reaction speed.

Distractions in the environment may also reduce participants' ability to concentrate, leading to delays in processing information and significantly increasing their average reaction time.



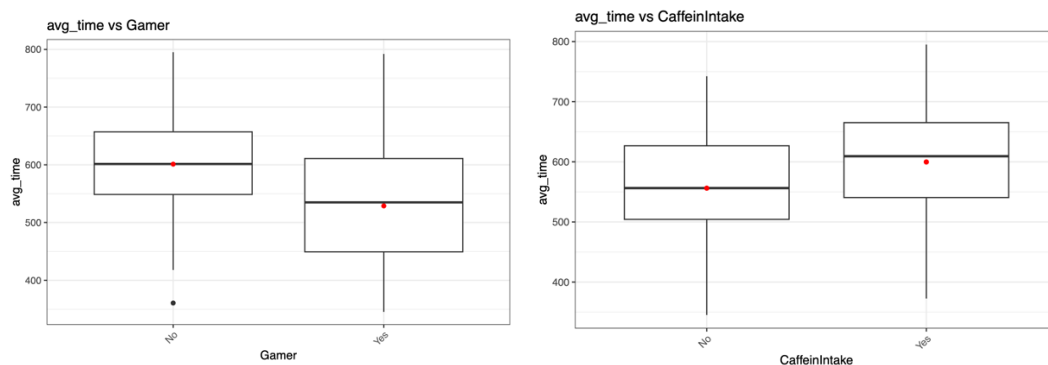
The medians of different temperatures are similar, which indicates that the avg_time may not be affected by environment temperature too much.

Overall, we noticed that the more experience someone had with shooting games, the faster their reaction time tended to be. There was a clear downward trend, with the “Advanced” group showing the lowest median reaction time. Even though the average for the “Expert” group was slightly higher than “Advanced,” it was still much faster compared to those with “None” or “Beginner” experience.



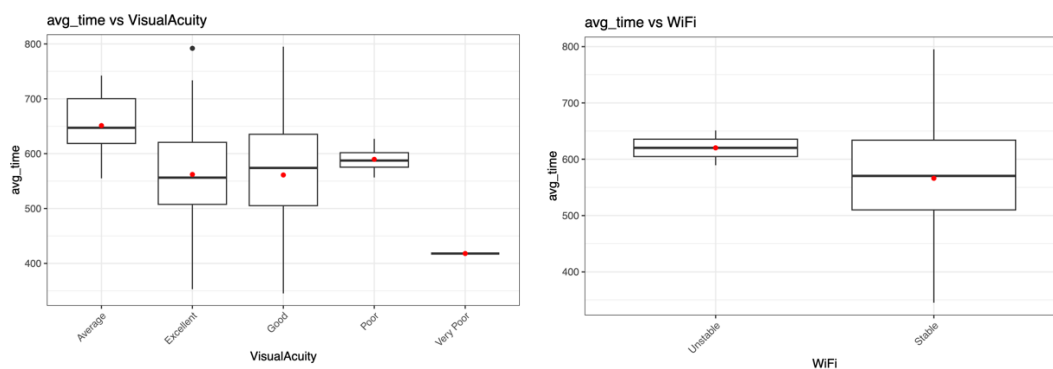
From the plot on the left, it looks like people who play video games often have much faster reaction times than those who don't play at all — even though some of them play every day. It's possible that playing every day could make them feel a bit tired, which might slightly slow down their reaction time. But overall, frequent gamers still respond faster than non-gamers.

As for how often people play sports, the pattern of reaction time looks more random and less consistent. This actually makes sense — because usually, people who really enjoy gaming might not exercise that much, but that doesn't necessarily mean their reaction time is worse.



From the first plot, we can clearly see that participants who play video games ("Yes") tend to have faster reaction times compared to those who don't ("No"). The median and mean values are both lower in the gamer group, suggesting better response speed. This supports the idea that regular gaming might help improve quick thinking and hand-eye coordination.

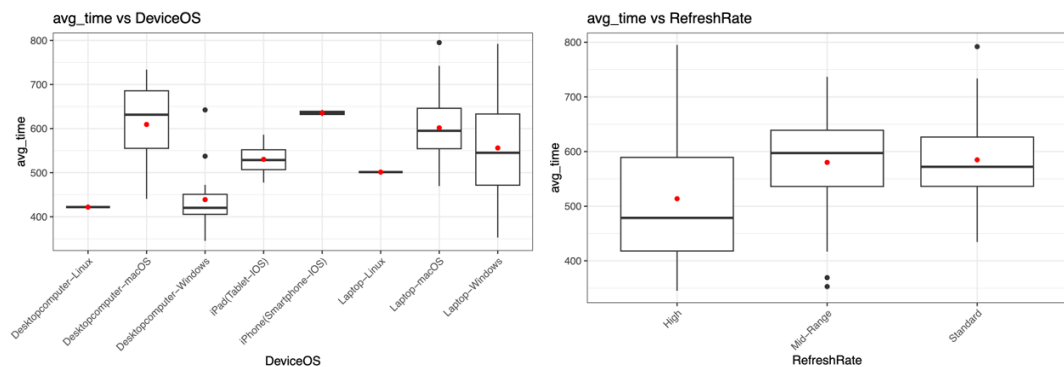
In the second plot, the average reaction time of people who consume caffeine is slightly higher than those who don't. This betrays our common sense. One possible explanation is that people who drink caffeine may already be feeling tired.



People who reported having "Very poor" vision had the fastest reaction time and this doesn't make sense. But overall, we still see a general pattern where better vision is linked to quicker reaction speed. People with "Good" or "Excellent" vision usually respond faster than those with worse eyesight.

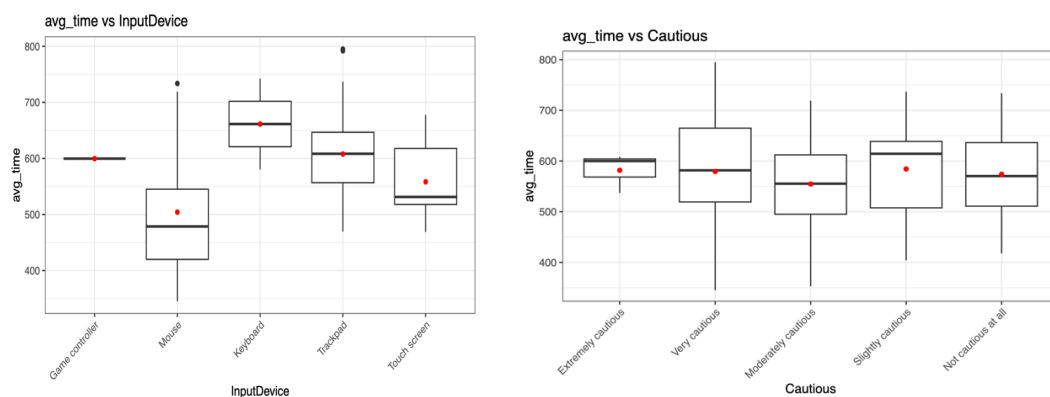
Also, having a stable WiFi connection seems to matter. Participants with good WiFi tend to have

shorter reaction times, which makes sense since unstable internet might cause delays during the online reaction test.



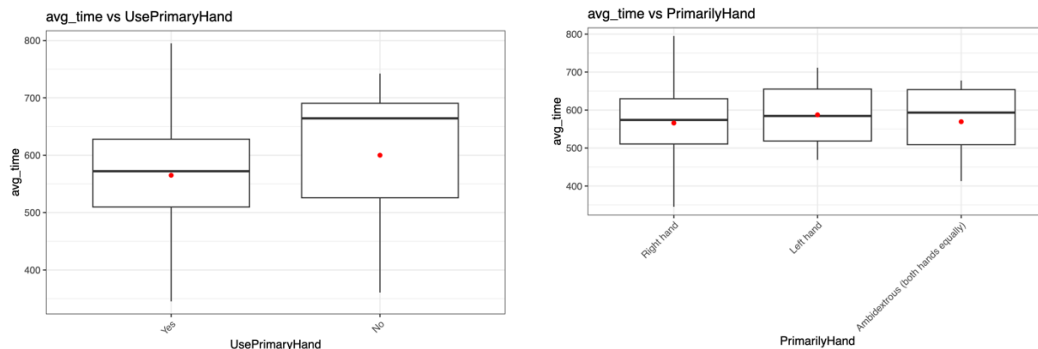
Overall, users with Windows desktops had the fastest reaction times compared to all other device types, even though there were two outliers. The sample sizes for Linux and iPhone users were too small, so it cannot suggest anything. In general, Windows devices performed better than Macs in terms of reaction speed.

As for the screen refresh rate, having a low or medium refresh rate didn't seem to make much difference in avg_time. However, people using high refresh rate displays showed significantly faster reaction times. At the same time, the variance in this group was also larger, suggesting more variation among those users.



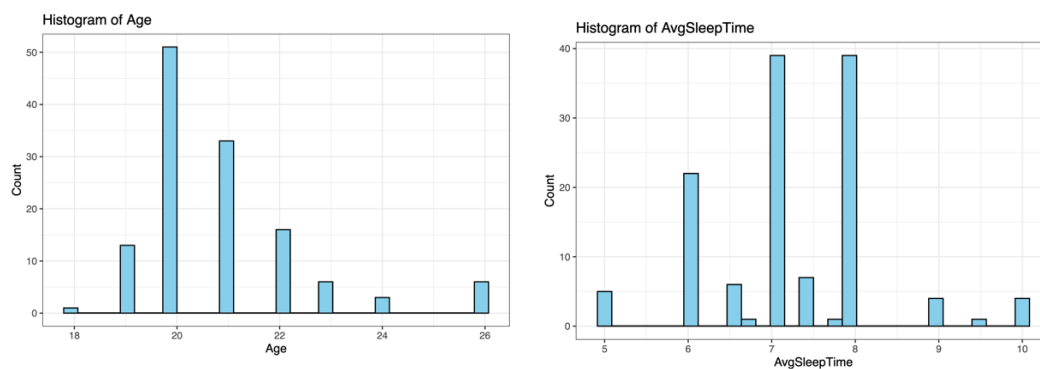
It's pretty clear that different input devices are related to different refresh rates. If we ignore the game controller group (since it had too few samples), the general order from fastest to slowest reaction time is: mouse > touch screen > trackpad > keyboard.

On the other hand, whether someone describes themselves as cautious or not doesn't seem to make a big difference in their average reaction time. There's no obvious pattern here.



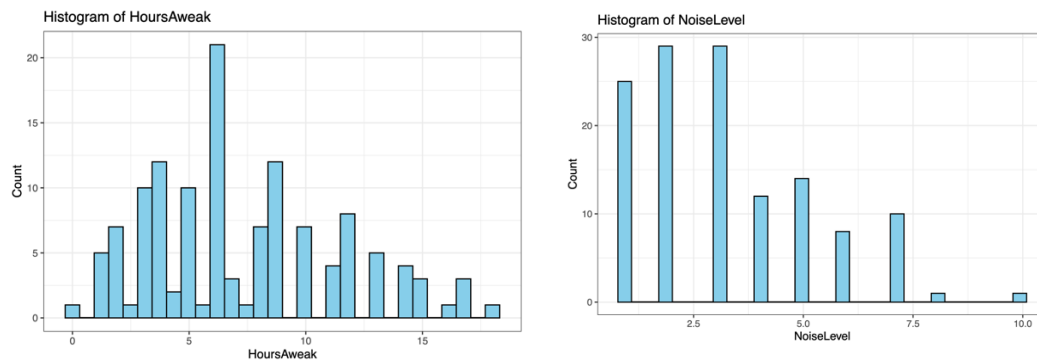
From the left plot, we can see that participants who didn't use their primary (dominant) hand during the test had overall higher average reaction times, and their results were more tightly clustered. In contrast, those who used their dominant hand showed more variation, but with a lower median reaction time. This suggests that using the non-dominant hand might slow down response speed.

However, the right plot shows that being left- or right-handed doesn't really make a big difference in reaction time. So, it's not about which hand is dominant — it's more about whether the person used their dominant hand or not during the test.



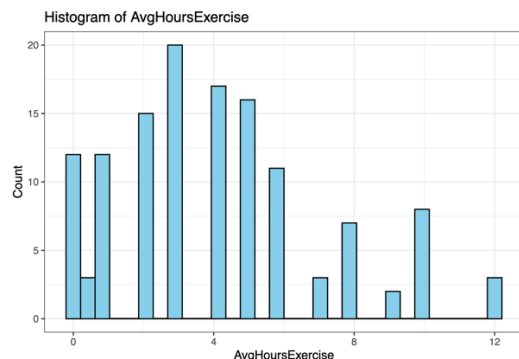
From the age histogram, we can see that most participants are around 20 years old, and the distribution is highly skewed. This shows that the sample mainly consists of young people, so the findings of this study are probably more representative of the college-age population.

Looking at the histogram of average sleep time, the data is tightly concentrated between 7 and 8 hours, with two clear peaks. This suggests that most participants are getting enough sleep. Of course, there are still quite a few samples with shorter or longer sleep durations, which means the dataset covers a wide range and has some diversity.



From the histogram of awake time (HoursAweek), most participants had been awake for 4 to 10 hours when they filled out the survey, with a noticeable peak around 6 hours. The distribution is also quite widespread. There are a few cases where participants were awake for over 15 hours, which might have affected their reaction times. Overall, the data shows a good variety.

As for the noise level (NoiseLevel), the histogram shows that most participants were in relatively quiet environments, with values mostly between 1 and 3. Very few people reported noise levels higher than 7, which means extremely noisy environments had a minimal impact on the overall sample.



From the histogram of AvgHoursExercise, we can see that most participants exercise around 2 to 5 hours per week, which shows a moderate level of physical activity. There are also some individuals who work out more than 8 hours weekly, creating a right-skewed distribution and indicating a small group of high-frequency exercisers. Then we applied a log transformation on AvgHoursExercise to reduce the skewness and make the variable more suitable for modeling.

Pre-processing:

Then we combined some variables' level and did some pre-processing..

Original Variable	Original Levels	New Variable	New Levels
Fatigue	5	Fatigue	3

Stress	5	Stress	3
NoiseLevel	10	Noise_level	3
VideoGamePlay	6	Game_Frequency	2
SportsPlay	6	Sports_Frequency	2
ShootingGames	5	Shooting_Levels	3
DeviceOS	8	Device + OS	4+3
First, second, third	Mean() -->	Avg_time	
AvgHoursExercise	Log(+1) -->	logAvgHoursExercise	

Model Building

Stepwise AIC:

To build an effective predictive model for average reaction time, we first used stepwise regression based on the AIC to select initial variables, resulting in a linear model with 27 predictors. Then, we conducted a VIF, and found *InputDeviceMouse* and *InputDeviceTrackpad* had VIF values 33.96 and 35.48. This indicates severe multicollinearity.

Therefore, we removed the *InputDevice* variable and built a new model (*AICmod2*) that included main effects along with first-order interaction terms involving *AvgSleepTime*. However, ANOVA results showed that all interaction terms had low F-values and p-values greater than 0.1 (e.g., *AvgSleepTime:Class*, $p = 0.815$; *AvgSleepTime:VisualAcuity*, $p = 0.825$), indicating that these interactions were not statistically significant. We then move the interaction term to the backend of the model and see if their significance changes (*AICmod3*). This new model confirmed that adding interactions did not lead to meaningful improvement. We then removed all interaction terms.

To further simplify the model and improve interpretability, we built a refined model (*AICmod4*) that kept the following variables: *Class*, *AvgSleepTime*, *CaffeinIntake*, *VisualAcuity*, *RefreshRate*, *Device*, *Fatigue_3*, *Game_Frequency*, and *UsePrimaryHand*. This model maintained reasonable predictive performance while reducing complexity. The adjusted residual sum of squares was 66.96. Key variables like *AvgSleepTime* ($p = 0.0023$), *CaffeinIntake* ($p = 0.0044$), *VisualAcuity* ($p < 0.05$), and *RefreshRate* ($p < 0.05$) were statistically significant, showing that they play meaningful roles in explaining average reaction time.

In summary, we selected *AICmod4* as the final model because it effectively explains the influence of key variables on reaction time while avoiding unnecessary complexity and multicollinearity issues.

Stepwise BIC:

In an alternative modeling pathway, we used the stepwise BIC for variable selection, resulting in a more concise initial model (*BICmod*) that included *Age*, *AvgSleepTime*, *InputDevice*, *Shooting_Levels*, and *Distraction*. While the overall model fit was moderate, we identified high multicollinearity in

InputDevice based on its VIF, leading to its removal.

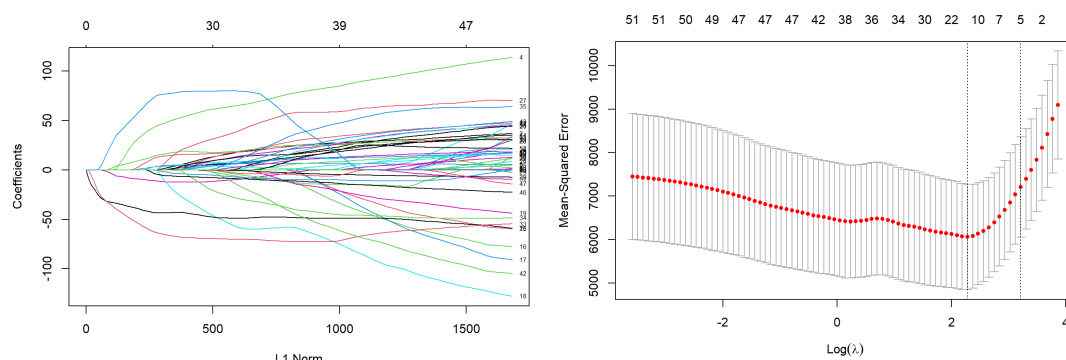
After removing *InputDevice*, we explored potential non-additive effects by constructing a new model (*BICmod2*) that included all main effects along with first-order interaction terms involving *Age* and *AvgSleepTime*. ANOVA analysis revealed that the interaction between *Age* and *Shooting_Levels* was statistically significant ($p < 0.05$), while other interactions such as *Age:Distraction* and *AvgSleepTime:Shooting_Levels* had high p -values and were not significant.

As a result, we retained only the significant interaction *Age:Shooting_Levels* and constructed a refined model (*BICmod3*) including *Age*, *AvgSleepTime*, *Shooting_Levels*, *Distraction*, and their interaction *Age:Shooting_Levels*. This model strikes a balance between simplicity and interpretability, highlighting meaningful interaction effects and offering a strong candidate for understanding key influences on reaction time.

Lasso:

We also applied the Lasso regression method to model average reaction time (*avg_time*), aiming to perform automatic variable selection among a large set of predictors. By applying L1 regularization to the regression coefficients, Lasso shrinks less important coefficients toward zero while retaining variables with strong explanatory power, thus producing a sparse and interpretable model.

Cross-validation results identified the optimal regularization parameter (λ), which effectively balanced model complexity and minimized mean squared error (as shown in the figure). The final Lasso model retained approximately 15 non-zero coefficients.

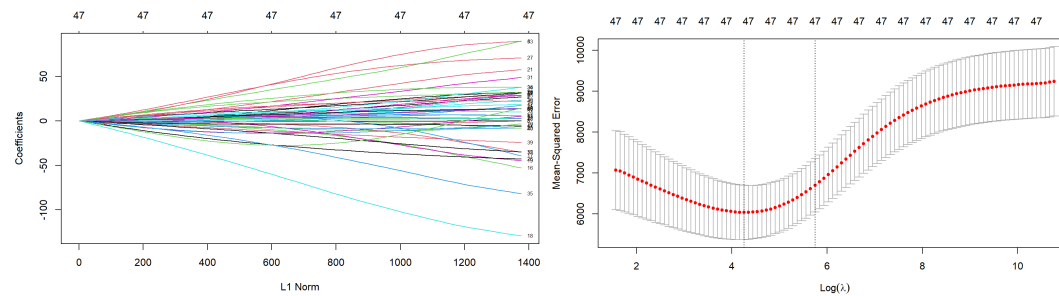


Ridge:

To further assess the stability of multivariable modeling and alleviate multicollinearity, we also applied Ridge regression to predict average reaction time. Unlike Lasso, Ridge does not shrink coefficients to zero. Instead, it retains all variables in the model while controlling the size of the coefficients to reflect their relative importance.

The final model included all predictors. Ridge regression thus provides a balanced approach, offering both predictive accuracy and interpretability, especially when the goal is to understand the

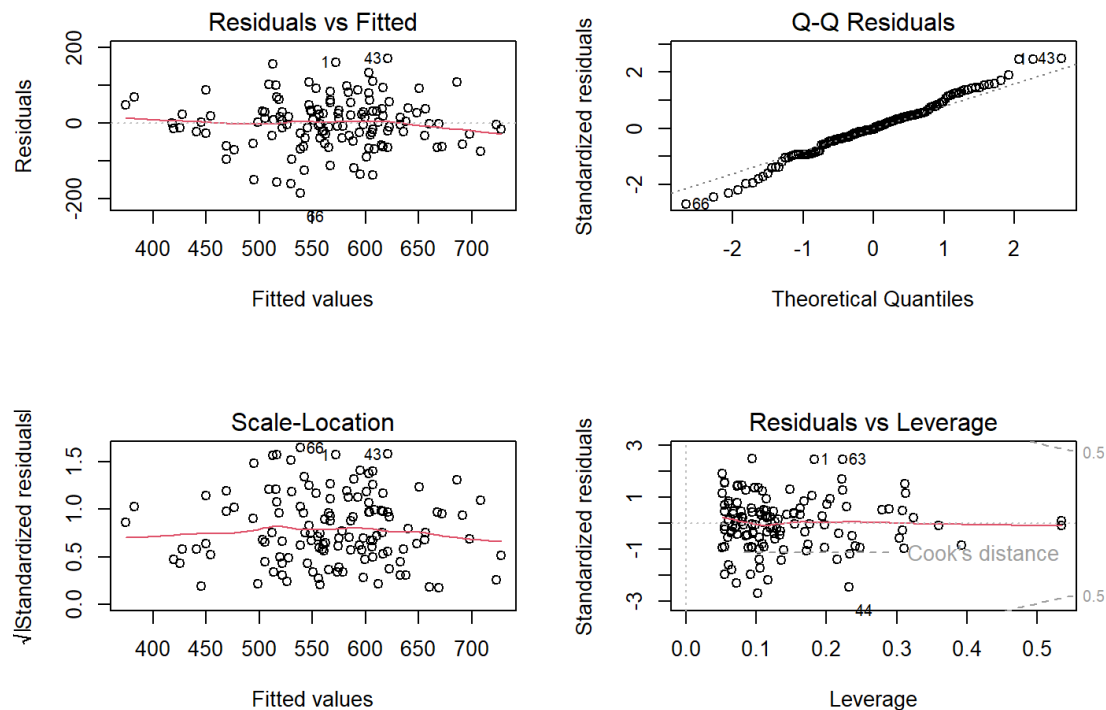
collective impact of all variables.



Model Diagnostics

AICmod4

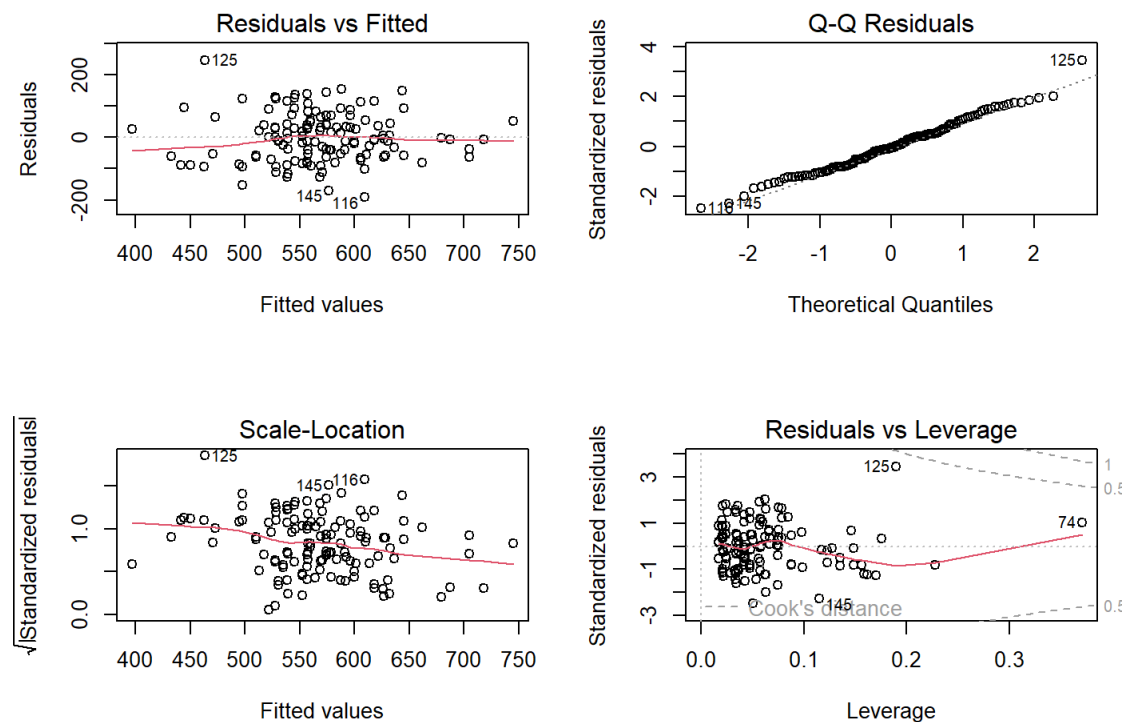
To make sure our final AIC model (AICmod4) was reliable, we ran some diagnostic checks. The residuals looked pretty evenly spread around zero in the Residuals vs Fitted plot, which means the model isn't obviously missing any patterns. Most of the points on Q-Q plot followed the diagonal line. We also checked whether the variance of the residuals stayed consistent across predictions. The Scale-Location plot didn't show any major fan shape, and the Breusch-Pagan test had a p-value just above 0.09, suggesting that heteroscedasticity isn't a big concern. Durbin-Watson test, and with a p-value of 0.82, there's no sign of autocorrelation issue. The Shapiro-Wilk test gave a p-value of 0.35, indicating there's no strong evidence that the residuals deviate from normality. The Cook's Distance also doesn't find any unusually influential points.



BICmod3

The Residuals vs Fitted plot does not show any clear systematic pattern since all points look like they are symmetric about residuals = 0. The Q-Q plot indicates a slight deviation in the lower tail, but can be seen as approximately normally distributed. This is further supported by the Shapiro-Wilk test ($p = 0.541$).

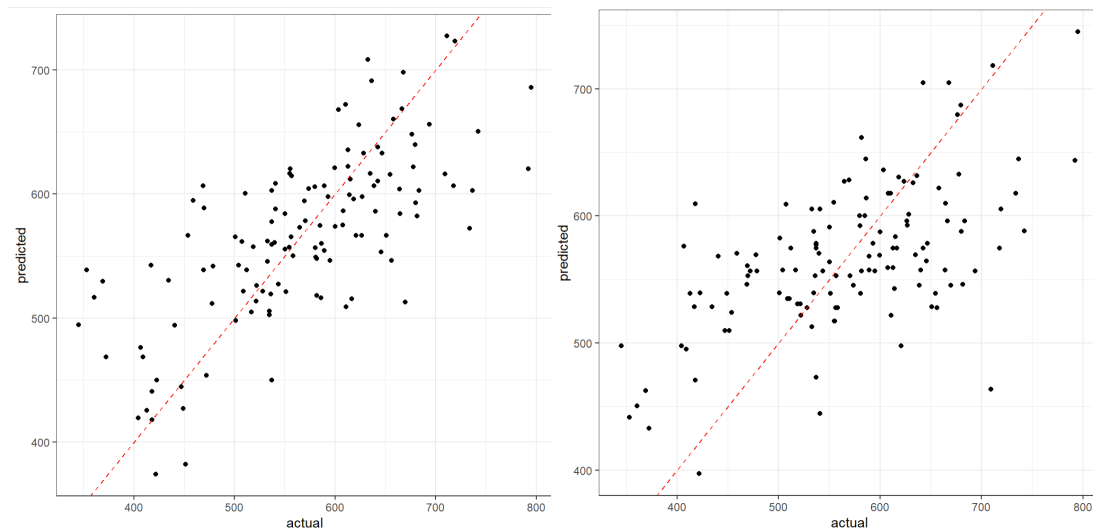
We then do Breusch-Pagan test suggests the presence of heteroscedasticity ($p = 0.010$). The Durbin-Watson test indicates some autocorrelation among residuals ($p = 0.021$). To address the issue of heteroscedasticity, we applied weighted least squares (WLS), using weights inversely proportional to the frequency of each Shooting_Levels group. The adjusted model performed well in the new Bp test with $p = 0.233$, showing that the weighting successfully reduced the problem of non-constant variance. The benefit of this BIC model is it only includes a small set of predictors. The final model we used was BICmod3.



We compared the models' prediction accuracy using RMSE for Ridge, Stepwise AIC, Lasso, and Stepwise BIC weighed regression model, and we get 62.51 ms, 66.96 ms, 70.01 ms, and 76.79 ms, respectively.

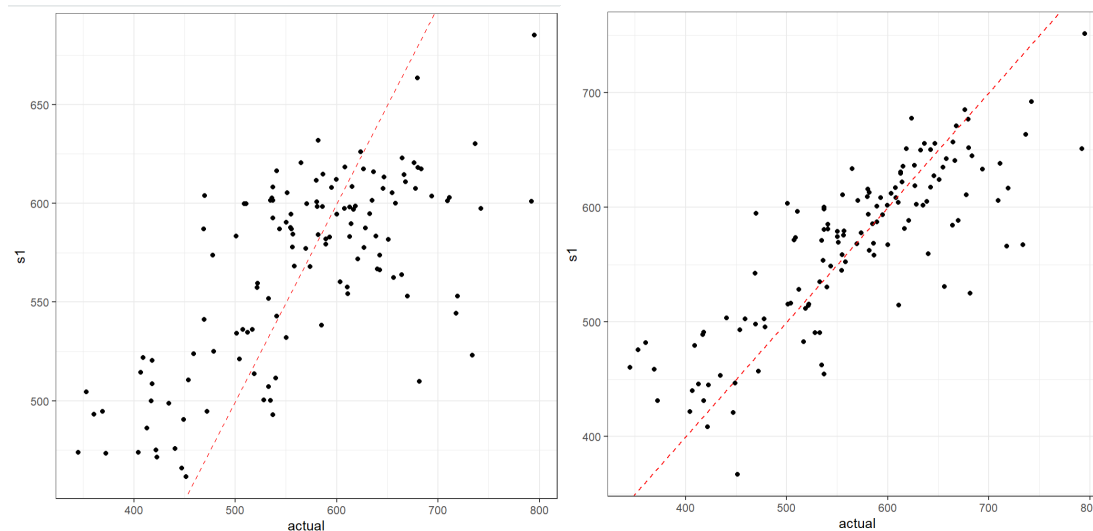
Ridge regression performed best, likely due to its ability to handle multicollinearity while preserving relevant features. The ridge has the lowest mse because it didn't delete any predictors. The AIC model has slightly higher error, possibly because it retained more predictors and tended to overfit. The Lasso model had a larger RMSE may because because some predictors are useful but were discarded. The BIC weighted regression model had the highest RMSE probaly because it excluded important predictors.

In conclusion, while the BIC model offers simplicity and control over variable selection, the Ridge regression model demonstrated the best prediction accuracy on this dataset.



AIC model Accuracy

Bic Model accuracy



Lasso Model Accuracy

Ridge Model Accuracy

Summary

This study explored what affects average reaction time, measured in milliseconds. We used survey data from UIUC students, which was cleaned and recoded before modeling. Several linear models were tested, including AIC and BIC stepwise regression, Lasso, and Ridge regression. We also ran diagnostic checks to compare their performance.

From a performance perspective:

1. The Ridge regression model achieved the lowest RMSE (53.7 ms), demonstrating strong robustness and generalization in a high-dimensional setting.

2. AIC and Lasso models showed similar performance, with the AIC model passing key diagnostic tests.
3. The BIC model was more conservative in variable selection and had higher RMSE, requiring Weighted Least Squares (WLS) correction to address heteroscedasticity.

Considering variable interpretability, statistical significance, and predictive accuracy, the Ridge regression model emerged as the best-performing method for modeling and predicting student reaction time.

One key challenge during modeling was variable selection. There are too many categorical variables with many levels. Grouping more levels with less levels is a challenging problem. In correct levels combining can lead to inaccurate variable selection. In the future, we plan to collect more data to make the model more stable, if is possible. Like the next 425 students can take the questionnaire again to see if our model has good prediction on their reaction time. We would also consider non-linear models or plotting interaction effects to get more accurate results.

We learned a lot in data cleaning process and learned how to group variables. We also learn how to use boxplot to see the trend of different categorical variables. We found that consume caffe may not make people respond quickly. A refreshing mind seems better. We suggest deleting some questions that look like they are overlapping, leading to multicollinearity issues. For example, primary hands can be deleted, and we can just remain using primary hand or not. The study and analysis may be fail if we have too many categorical variable instead of numerical measurement, since make of them are just people's feeling.